# A Complex Namespace for the rho Calculus

Paired messages, two quotations, and an intertwining COMM

L.G. Meredith
F1R3FLY.io

Version 2 — a record of the core calculus

**Abstract**

This note records, in one place, the core of a variant of the rho calculus designed so that the namespace carries the structure of a complex scalar field rather than that of a free commutative monoid. Three changes are made to the term language and one to the reduction relation. Messages carry *pairs* of processes, so that a message is a formal difference and negation is structural. Names are quotations of pairs, so that negation is an operation on *addresses* and not merely a decoration on sends. There are two quotation operators and, of necessity, two dropping operators; the two quotations are read as a real and an imaginary sector. Communication *intertwines* the sectors, so that a communication event is multiplication by the imaginary unit, and communication between participants of opposite polarity exchanges the components of the received pair, so that opposite polarity contributes the sign. The note states what has been settled, states the design forks that have not, and enumerates the obligations the design incurs. It also records the evaluation product against which the eventual pairing is defined, together with the pole and the residuation that give the dagger, taking the pole to be the zero process and the perp to be residuation to it. It proves almost nothing. Its purpose is to fix notation and to make the open questions precise enough to be worked on.

## Contents

# 1 What this note is

## 1.1 Scope

The present document is a *record*, not an argument. It writes down the term language, the structural theory and the reduction relation of a variant calculus that has emerged from a sequence of design decisions, together with the reasons each piece is present. It does not develop the semantics that motivated the design; that semantics — an interpretation of the operations of quantum mechanics over a reflective process calculus, with processes as vectors, names as scalars, evaluated contexts as covectors and contexts as operators — is the subject of separate work, and is recalled here only far enough to explain why each feature was added.

Two things are deliberately excluded. The note gives no metatheory: no subject reduction, no congruence result, no decidability. And it gives no semantics: no interpretation map, no inner product, no dagger beyond the one line needed to say which operation the dagger is built from. Both are named as obligations in Section 10.

## 1.2 Status markers

Throughout, claims carry one of three markers.

**SETTLED** The design decision has been taken and the consequences worked through. It may still be wrong, but it is not open.

**FORK** Two or more readings are consistent with what has been settled, and the choice has not been made. The alternatives are stated.

**OPEN** An obligation the design incurs and does not discharge.

### 1.3 Revision note

Version 1 of this note had receipts bind a pair of name variables and stated the communication rule in two clauses, one per relative polarity. Both features are withdrawn. Negation on names lifts uniquely to processes (Proposition 7), the whole received pair is deposited into a single bound variable, and the polarity clause becomes a factor in one rule. The change is not cosmetic: requiring reduction to be equivariant for the lifted negation *selects* the deposit convention, and version 1's convention fails that test (Remark 26). The surviving pair sort occurs only on messages and inside quotations.

### 1.4 Notational rules

The calculus is called the rho calculus and is never written with the Greek letter that follows pi; the name abbreviates *reflective higher order*, and the transliteration is the pun. Quotation is written @ and dropping $*$; corner quotes are not used.

The design discussions that produced this calculus wrote the pair constructor as an infix asterisk, $Q_p * Q_n$. That notation collides with the dropping operator, so pairs are written here with angle brackets, $\langle Q_p, Q_n \rangle$, and the reader translating from those discussions should read $\langle Q_p, Q_n \rangle$ wherever $Q_p * Q_n$ appears there.

## 2 Why the namespace is expanded

The calculus below is the minimal response to four defects, each of which is a defect of the *unexpanded* namespace. They are recalled because each feature of the design is answerable to one of them, and a reader who forgets the defects will find the features arbitrary.

**The scalars have no additive inverses.** In the ordinary rho calculus, structural congruence is associativity, commutativity and a unit for parallel composition. Consequently $\mathsf{Proc}/\equiv$ is the free commutative monoid on the non-parallel components of processes, and $\mathsf{Name}$, being $\mathsf{Proc}/\equiv$ transported through quotation, is that same free commutative monoid. A free commutative monoid has no inverses and no cancellation. Whatever else it can express, it cannot express the vanishing of a sum of nonzero contributions. Since destructive interference is exactly such a vanishing, no coefficient structure carried by names can produce it.

**The multiplicative and additive units collide.** Name multiplication is $@P \cdot @Q := @(P \mid Q)$, whose unit is $@\mathsf{0}$. Any addition on names built from summation also has $@\mathsf{0}$ for its unit. The rig law $0 \cdot x = 0$ then fails, since $@\mathsf{0} \cdot x = x$. There is no absorbing element, and hence no rig.

**There is no conjugation.** Parallel composition is commutative, so any pairing built from it is symmetric rather than Hermitian, and there is no involution available on the coefficients to repair this. An involution must come from somewhere; the unexpanded calculus has none of order two acting on scalars.

**Multiplication by a monomial is a shift, not a scaling.** This is the sharpest of the four. In the free commutative monoid, multiplying a name by another name is injective and never decreases anything; it permutes basis elements rather than combining them. Any attempt to realise the imaginary unit as an ordinary name — for instance by adjoining a ground process $1$ with $1 \mid 1 \mid 1 \mid 1 \equiv 0$, so that $@1$ has order four — therefore fails for a reason that is fatal rather than technical: $@R$ and $@(1 \mid R)$ are names of *different addresses*, so no receiver hears both and nothing superposes. Multiplication by $i$ must move a message *between sectors at one address*, not between addresses. This is the requirement that forces the sector to be a coefficient attached to a name rather than a factor inside it, and it is the reason the design expands the namespace instead of enriching the process language.

*Remark* 1. The four defects are not independent. Repairing the first by group completion supplies the involution needed for the third, and separates the units demanded by the second. Only the fourth is a genuinely separate requirement, and it is the one that dictates *where* the new structure is placed.

# 3 Syntax

## 3.1 Sorts

The calculus has three sorts: processes, pairs and names. Pairs are the new sort. A pair is intended as a formal difference of processes, its first component positive and its second negative. Pairs occur in two places only, on messages and inside quotations; binders do not mention them, for the reason given in Remark 24.

**Definition 2** (Grammar)**.**

$$\begin{aligned} P, Q &::= 0 \mid P \mid Q \mid \mathsf{for}(y \leftarrow x)P \mid x!A \mid *_l x \mid *_r x \\ A, B &::= \langle P, Q \rangle \\ x, y &::= @_l A \mid @_r A \end{aligned}$$

Three points of reading. A send carries a pair, not a process. A receipt binds a single name variable, and the negative component is reached from it by $\ominus$, which Proposition 7 makes available on every term. And a name is a quotation of a pair, tagged with a sector $l$ or $r$.

**Convention 3** (Drops in process position)**.** $*_s x$ is stipulated above to be a process, whereas the pair quoted by $x$ has two components. We take $*_s x$ to denote the *positive* component of the $s$-sector content of $x$, so that $*_s(\ominus x)$ is the negative component. No generality is lost and the ordinary reading of $*$ is preserved.

**Convention 4** (Abbreviations)**.** We write $x!Q$ for $x!\langle Q, 0\rangle$. Together with Convention 3 and the choice $@ := @_l$, $* := *_l$, this is what makes the ordinary $\mathsf{rho}$ calculus a sub-language; see Section 7.

## 3.2 Negation

**Definition 5** (Negation on pairs and names)**.** $\ominus$ is defined on pairs and on names by component exchange at the top:

$$\ominus\langle P, Q \rangle := \langle Q, P \rangle, \qquad \ominus @_s A := @_s(\ominus A) \quad (s \in \{l, r\}).$$

The definition does not recurse into the components.

**Proposition 6** (Negation is a total involution on names)**.** *$\ominus$ is defined on every name without reference to reduction, and $\ominus\ominus x = x$.*

*Proof.* Immediate from component exchange being an involution on ordered pairs. $\square$

**Proposition 7** (Unique lifting to processes)**.** *SETTLED. There is exactly one extension of $\ominus$ from names to processes which is a homomorphism of the term algebra, namely*

$$\ominus \mathsf{0} := \mathsf{0}, \quad \ominus(P \mid Q) := \ominus P \mid \ominus Q, \quad \ominus(x!\langle P, Q\rangle) := (\ominus x)!\langle \ominus P, \ominus Q\rangle,$$

$$\ominus(\mathsf{for}(y \leftarrow x)P) := \mathsf{for}(y \leftarrow \ominus x) \ominus P, \quad \ominus(*_s x) := *_s(\ominus x).$$

*It satisfies $\ominus\ominus P = P$, and it descends to $\equiv$ and to $\equiv_N$.*

*Proof.* Processes are freely generated by the term formers over names, so a map on names extends uniquely to an endomorphism of the term algebra; the displayed clauses are that extension. Involutivity follows by induction from Proposition 6. For $\equiv$, the clauses are homomorphic and leave the binder alone, so associativity, commutativity and the unit are preserved. For $\equiv_N$, applying $\ominus$ to the quote-drop law (Definition 19) at $x$ yields the quote-drop law at $\ominus x$, since $*_s(\ominus x)$ is the negative component of $x$. $\square$

*Remark* 8 (What the lift is, and is not). It is worth being explicit, because the notation invites a conflation that the rest of the note depends on avoiding. The lifted $\ominus$ is an *automorphism of the term algebra* — a relabelling of the namespace — and not an additive inverse. In particular $P \mid \ominus P$ is inert and is not $\mathsf{0}$: a send and its mirror have no receipt between them, so nothing reduces. Cancellation does not come from $\ominus$ on processes; it comes from the completion equations on pairs, and orthogonality accordingly comes from where Remark 53 says it does.

Homomorphism for $\mid$ is therefore a *consequence* of freeness, not a design decision, and it carries none of the algebraic content it would carry if $\ominus$ were an inverse.

*Remark* 9 (Why the pair is inside the quotation). SETTLED. If pairs occurred only on messages, then names would quote single processes and $\ominus x$ would be undefined: negation would be an operation on messages that the namespace cannot express, so $x$ and $\ominus x$ would not both be addresses, the polarity of a participant in a communication could not be stated, and Proposition 7 would have nothing to lift. Placing the pair inside the quotation makes negation an operation on addresses and lets the reduction rule mention $\ominus x$ as a name like any other.

There is a second reason, of a different kind. Under the reading in which parallel composition is addition on processes and name multiplication is $@P \cdot @Q = @(P \mid Q)$, quotation is an exponential. Completing to formal differences *before* exponentiating gives a namespace that is the exponential image of a group; completing afterwards does not, and that is why earlier attempts to obtain negation dynamically, as a consequence of annihilation, kept having to produce the sign at reduction time rather than having it available in the syntax.

# 4 Sectors, polarity, and the group of a name

## 4.1 Two quotations force two drops

**Proposition 10** (Collapse)**.** *Suppose the calculus has two quotation operators $@_l, @_r$ and a* single *dropping operator $*$ satisfying $*(@_s A) = A$ for both $s$, together with the quote-drop law $@_s(*x) \equiv_N x$. Then $@_l A \equiv_N @_r A$ for every $A$.*

*Proof.* Instantiate the quote-drop law at $x := @_r A$ and $s := l$: $@_l A = @_l(*(@_r A)) \equiv_N @_r A$. $\square$

So the two quotations are distinguishable only if the drops are split as well; a map has at most one two-sided inverse. This is why Definition 2 carries $*_l$ and $*_r$ and not a single $*$.

**Definition 11** (Sector projection)**.** $*_s(@_t A) := A$ if $s = t$, and $\langle 0, 0\rangle$ if $s \neq t$.

*Remark* 12. FORK. Definition 11 makes the drops total, so that $*_l$ and $*_r$ behave as real and imaginary part. The alternative — leaving the cross cases stuck — makes the drops partial and puts the sector distinction outside the equational theory, where the matcher cannot use it. Totality is taken here, but the choice interacts with Obligation 9 below and is not innocent.

## 4.2 The four names of an address

Each name has an underlying pair and two binary attributes: its sector, $l$ or $r$, and its polarity, which is the choice between a pair and its exchange. Fixing the underlying pair up to exchange therefore leaves four names.

**Definition 13** (Address and phase)**.** For $x = @_s A$, write $\mathsf{sec}(x) := s$ and let $\mathsf{addr}(x)$ be the pair $A$ taken up to $\ominus$. The four names sharing an address are

$$@_l A, \qquad @_r A, \qquad @_l(\ominus A), \qquad @_r(\ominus A).$$

**Definition 14** (The imaginary unit)**.** SETTLED, with the caveat of Remark 16. Multiplication by $i$ is the successor map on the cycle

$$@_l A \longmapsto @_r A \longmapsto @_l(\ominus A) \longmapsto @_r(\ominus A) \longmapsto @_l A.$$

Equivalently: passing from $l$ to $r$ leaves the pair alone, and passing from $r$ to $l$ exchanges it.

**Proposition 15.** *The map of Definition 14 has order four, its square is $\ominus$, and the four names of an address form a torsor under $\mathbb{Z}/4$ with $\mathsf{ph}(-)$ the resulting coordinate, $\mathsf{ph}(@_l A) = 1$.*

*Proof.* Two applications carry $@_l A$ to $@_l(\ominus A)$, which is $\ominus$ by Definition 5; $\ominus$ is an involution by Proposition 6, so the map has order four. The action is free and transitive on the four names by Definition 13. $\square$

*Remark* 16 (Why $i$ is not the composite of two independent involutions)*.* This is the most delicate point in the design and it must not be glossed. Sector exchange and pair exchange are each involutions, and if they are taken as *independent* the group they generate is the Klein four-group. The Klein four-group is the group of the split-complex numbers, in which $j^2 = +1$ and there are zero divisors, $(1+j)(1-j) = 0$. Zero divisors are fatal here, since a pairing would then vanish for reasons having nothing to do with orthogonality. Definition 14 is precisely the stipulation that the two exchanges are *not* independent: the descending passage from $r$ to $l$ carries the pair exchange with it, so that the generated group is cyclic of order four rather than Klein. The asymmetry is not decoration. It is the content of the requirement $i^2 = -1$, and it is what a design with two free involutions cannot supply.

*Remark* 17 (Only one doubling)*.* Name multiplication is parallel composition, hence commutative. The Cayley–Dickson construction preserves commutativity once and loses it at the second step. A namespace whose product is parallel composition therefore admits exactly one doubling and no more: the calculus can be complex, and cannot be quaternionic, for a structural reason rather than by stipulation.

# 5 Structural theory

**Definition 18** (Structural congruence)**.** $\equiv$ is the least congruence containing alpha-equivalence and satisfying associativity, commutativity and $\mathsf{0}$ as unit for $|$, together with

$$\ominus \ominus A \equiv A.$$

**Definition 19** (Name equivalence)**.** $\equiv_N$ is the least congruence satisfying

$$@_s\langle *_s x, \ominus *_s x\rangle \equiv_N x \quad \text{when } \mathsf{sec}(x) = s, \qquad \frac{A \equiv B}{@_s A \equiv_N @_s B}.$$

*Remark* 20 (The completion equations, and where they live). FORK, and the most consequential one in the note. A pair is intended as a formal difference, which suggests two further equations:

$$\langle Q, Q\rangle \equiv \mathsf{0}, \qquad \langle P, Q\rangle \equiv \langle P \mid R, Q \mid R\rangle.$$

These are what make the pair sort the group completion of processes rather than merely an ordered pair. But $\equiv_N$ contains $\equiv$ under quotation, and COMM fires on $\equiv_N$. Imposing them therefore makes them equations *on addresses*, which the matcher must decide.

The two readings differ sharply.

- **Arithmetic names.** The completion equations hold on names. Cancellation is then real, $\ominus$ is genuine negation, and destructive interference is the matcher discovering that a pair's components have become congruent. The cost is that unique factorisation is lost, the monomial basis with it, and the decidability of name matching becomes a serious question.
- **Coloured names.** The completion equations do not hold on names, which are then a free set with a $\mathbb{Z}/4$ action. Negation is a label carried along and read by the reduction rule, exactly as the sector is. The matcher is untouched; the arithmetic is deferred to a coefficient layer outside the calculus.

The reduction rule of Section 6 never asks whether two differently-signed names are equal, only whether they share an address, so the coloured reading suffices for the operational semantics as it stands. The arithmetic reading is what the semantic programme wants. The note takes neither; it records that the choice determines whether cancellation is a structural event or an external one.

# 6 Reduction

## 6.1 Matching is sector-blind and polarity-blind

**Definition 21** (Address matching)**.** SETTLED. A receipt at $x$ and a send at $x'$ are *matched* when $\mathsf{addr}(x) = \mathsf{addr}(x')$, irrespective of $\mathsf{sec}(x), \mathsf{sec}(x')$ and irrespective of polarity.

This is forced by the fourth defect of Section 2. If $@_l A$ and $@_r A$ were distinct addresses, then $@_l A!Q \mid @_r A!Q$ would be two messages that no single receiver can hear, rather than a superposition of one message with itself in two sectors; nothing would ever combine, the sector would be a superselection rule, and the doubling would carry no information. Sector and polarity are coefficients on a name, not part of its identity.

### 6.2 The rule

**Definition 22** (Relative polarity)**.** For matched $x$ and $x'$, the *relative polarity* $\delta(x, x')$ is the identity when $x' = x$ and $\ominus$ when $x' = \ominus x$. It is well defined by Definition 13, since matched names differ by an element of the $\mathbb{Z}/4$ torsor and polarity is the component of that element which Definition 14 squares to.

**Definition 23** (COMM)**.** Let $t := \mathsf{sec}(x)$ and let $\mathsf{op}(l) := r$, $\mathsf{op}(r) := l$. For matched $x, x'$ in the sense of Definition 21,

$$\mathsf{for}(y \leftarrow x)P \;\mid\; x'!A \;\to\; P\{\, @_{\mathsf{op}(t)}\big(\delta(x, x')\, A\big) \,/\, y \,\}$$

The rule is closed under parallel composition and under $\equiv$ in the usual way.

One clause, two effects, and they remain independent of one another.

**Sector: communication intertwines.** SETTLED. The quotation deposited is in the sector *opposite* to the one the receipt listened on. A communication on the real axis deposits an imaginary name; a communication on the imaginary axis deposits a real one. So a communication event is the map $l \mapsto r \mapsto l$, and by Definition 14 it is multiplication by $i$.

**Polarity: opposition exchanges the pair.** SETTLED. When the participants have opposite polarity, $\delta$ is $\ominus$ and the deposited pair is exchanged, which by Definition 5 is multiplication by $\ominus$. So opposite polarity contributes the sign, and $i^2 = \ominus 1$ is realised as: two sector rotations equal one polarity exchange.

*Remark* 24 (One binder, one clause)*.* SETTLED, and the reason the grammar of Definition 2 binds a single variable. An earlier presentation of this calculus had receipts bind a pair $\langle y_p, y_n \rangle$ and stated COMM in two clauses, one per relative polarity, each depositing $@\langle Q_p, \mathsf{0} \rangle$ and $@\langle Q_n, \mathsf{0} \rangle$ into the two variables. Both features are removed by Proposition 7. The whole received pair goes to the single variable $y$, and the negative component is $*_s(\ominus y)$, so the second binder was never carrying information the first did not. The polarity clause then ceases to be a second case and becomes the factor $\delta$, which is the statement that the relative group element of the two participants is the phase deposited — visible in the rule rather than distributed over two of them.

**Proposition 25** (Equivariance)**.** *$P \to P'$ implies $\ominus P \to \ominus P'$.*

*Proof.* $\ominus$ is a homomorphism leaving binders alone (Proposition 7), so it carries a redex $\mathsf{for}(y \leftarrow x)P \mid x'!A$ to $\mathsf{for}(y \leftarrow \ominus x) \ominus P \mid (\ominus x')!(\ominus A)$, which is a redex: matching is polarity-blind (Definition 21) and $\delta(\ominus x, \ominus x') = \delta(x, x')$. Both reducts are $\ominus P\{@_{\mathsf{op}(t)}(\delta(x, x') \ominus A)/y\}$, using $\ominus @_s B = @_s(\ominus B)$ and the commutation of $\ominus$ with substitution, which holds because $\ominus$ acts uniformly on the names of $P$. □

*Remark* 26 (Equivariance is a selection principle)*.* Proposition 25 is not decoration; it is what picks the deposit convention out of the alternatives. Requiring reduction to commute with the unique lifting of Proposition 7 rules out the earlier convention, under which the deposited name was always $@_{\mathsf{op}(t)}\langle Q, \mathsf{0} \rangle$ with the payload in the positive slot: nothing of that shape lies in the image of an equivariant rule, since $\ominus @\langle Q, \mathsf{0} \rangle = @\langle \mathsf{0}, Q \rangle$, and the failure is uniform at one application of $\ominus$ per communication. Depositing the whole pair is the unique repair, and it is the convention of Definition 23. That the same convention also collapses the pair binder and the second COMM clause is the reason to prefer it on grounds other than symmetry.

*Remark* 27 (Phase is path length). An immediate consequence of the sector effect. The sector of a name is the parity of the number of communications that produced it, so the relative phase between two derivations reaching a common outcome is the parity of the difference of their lengths. This is worth naming, because the question of where destructive interference could come from in an unmodified rho calculus had the answer *length-asymmetric reconvergence*, with no mechanism to supply it: concurrency diamonds are symmetric, so the holonomy vanishes. Here the phase *is* the length asymmetry, with no cost functor, no weight map and no external coefficient algebra.

*Remark* 28 (The identity step is not fictitious). Under the reading of Remark 27, a step that changes no process still rotates the sector. A device introduced elsewhere to complete deficient rows, and justified there case by case, acquires an independent justification here: doing nothing and doing nothing once are genuinely different.

### 6.3 The four-case refinement

*Remark* 29. FORK. Definition 23 computes the deposited sector from the receipt's channel alone. But Definition 21 makes matching sector-blind, so a receipt at $@_l A$ may be served by a send at $@_r A$, and the rule as stated ignores the sender's sector. The natural completion is that the two sectors *multiply*:

| | send $l$ | send $r$ |
|---|---|---|
| recv $l$ | $1$ | $i$ |
| recv $r$ | $i$ | $-1$ |

with $-1$ realised as $\ominus$ on the deposited pair and the off-diagonal entries being exactly the intertwining of Definition 23. Under this reading the rule of Definition 23 is the off-diagonal half of the table, and $i^2 = -1$ is a case of the reduction rule rather than a separate stipulation.

The refinement is attractive and is not adopted here, because it changes what the rule computes rather than extending it: on the diagonal it deposits a name in the sector it received on, which contradicts the plain reading of "communication intertwines". Settling this is the first item of Section 10.

*Remark* 30 (Amplitude is multiplicity). Under Definition 21, the expression $@_l A!Q \mid @_r A!Q$ is a single address carrying one copy in each sector, and with the pair supplying signs this is $(1+i)$ times a send at $A$. Generally, the coefficient of a message is the multiset of sectors and polarities its copies occupy, so Gaussian-integer coefficients are realised as multiplicity under parallel composition, with no external coefficient algebra and no tagging. This is the payoff that motivated Definition 21, and it should be the first thing checked against any proposed revision of the rule.

## 7 Conservativity

**Proposition 31** (Embedding). *FORK, stated as a target rather than a theorem. Let $\iota(-)$ send an ordinary rho term to the term of Definition 2 obtained by*

$$\iota(x!Q) := \iota(x)!\langle \iota(Q), \mathsf{0}\rangle, \qquad \iota(@P) := @_l\langle \iota(P), \mathsf{0}\rangle,$$
$$\iota(\mathsf{for}(y \leftarrow x)P) := \mathsf{for}(y \leftarrow \iota(x))\iota(P), \qquad \iota(*x) := *_l \iota(x)$$

*homomorphically elsewhere. Then $\iota(-)$ is injective and preserves $\equiv$.*

*Remark* 32 (Conservativity fails for reduction, and deliberately)*.* The image of $\iota(-)$ is not closed under $\to$. A single communication moves a name off the real axis, so an ordinary $\mathsf{rho}$ program is not, in this calculus, a program that stays classical. This is not an oversight to be repaired: it is the content of Remark 27, since a calculus in which ordinary programs remained on the real axis would be one in which no phase ever accumulated. The design is a genuinely different calculus, not an extension, and it should be presented as such.

If a conservative reduction is wanted, the diagonal entries of the table in Section 6.3 supply it: under that reading, a real receipt served by a real send stays real, and the ordinary calculus embeds as the real sub-calculus. That is an argument for the refinement, and is recorded here rather than in Section 6.3 because it is the strongest one.

# 8 Evaluation and residuation

The calculus above is a term language and a reduction relation. The construction it serves needs two further operations: an evaluation, which runs a composite to its terminal states and collects them, and a pole, against which the dagger is obtained by residuation. Both are recorded here. Following the design decision that the pole be taken as simply as possible for the time being, $\perp$ is the zero process and $P^{\perp}$ is residuation to it — the linear perp.

## 8.1 Inert states and evaluation

**Definition 33** (Inert)**.** $P$ is *inert* when $P \not\to$; that is, when no COMM applies to any of its components under $\equiv$. Write $\mathsf{Inert}$ for the inert processes.

Inertness is not triviality: $\mathsf{0}$ is inert, and so is $x!A$ with no matching receipt. The distinction matters throughout what follows, and is the reason $\perp$ below is not simply "is inert".

**Definition 34** (Evaluation product)**.**

$$P \odot Q \;:=\; \sum_{\substack{R \in \mathsf{Inert} \\ P|Q \to^* R}} R$$

where the sum ranges over terminal states with multiplicity, one summand per *causal history* ending at $R$.

**Proposition 35** (Commutativity)**.** *$P \odot Q \equiv Q \odot P$.*

*Proof.* $|$ is commutative and the index of the sum mentions $P$ and $Q$ only through $P \mid Q$. ☐

**Definition 36** (Evaluation)**.** $\Downarrow P := \mathsf{0} \odot P$.

*Remark* 37 (Histories, not paths). SETTLED, and load-bearing rather than hygienic. Indexing the sum by reduction *sequences* rather than by causal histories counts the linearisations of a partial order, so $m$ independent redexes contribute a spurious factor of $m!$. The multiplicity in Definition 34 is therefore the number of histories, equivalently the number of maximal chains in the reduction order modulo permutation of independent steps. Concurrency squares are filled; what is summed is what survives the quotient.

*Remark* 38 (What $\odot$ is not). Three properties fail, and each failure is informative rather than repairable.

*No unit.* $\mathsf{0} \odot P =\Downarrow P$, which is $P$ only when $P$ is already inert. So $\mathsf{0}$ is a unit on $\mathsf{Inert}$ and nowhere else, which is one reason the carrier of the eventual algebra must be cut down — see Section 8.2.

*Not idempotent.* $P \odot P \neq P$ in general, because two copies of a process interact. This is a feature: it is what makes multiplicity meaningful, and it is why the additive structure cannot be mixed choice, whose idempotence would force $2 = 1$.

*Not associative.* $(P \odot Q) \odot S$ forces $P \mid Q$ to reach a terminal state before $S$ arrives, so any global reduction of $P \mid Q \mid S$ that schedules a communication between $P$ and $S$ early, destroying a $P$–$Q$ redex, is a history of the unbracketed composite and of no bracketed one. The bracketed product undercounts, and the deficit is exactly the cross-redexes. Writing $M_R$ for the operator $\square \mid R$ and $T$ for one-step reduction, the deficit is the commutator $[T, M_R]$: the redexes between $R$ and the state that existed in neither alone. $M_R$ alone is diagonal in the occupation basis and commutes with every $M_S$; $T$ alone is the off-diagonal part; neither is interesting, and their failure to commute is where the content of the construction lives.

**Convention 39** (The $n$-ary form)**.** Because of the last item, the primitive is not the binary product but the family of symmetric $n$-ary evaluations

$$\odot(R_1, \ldots, R_n) \;:=\; \sum_{\substack{R \in \mathsf{Inert} \\ R_1 | \cdots | R_n \to^* R}} R$$

with one global evaluation and no bracketing. Symmetry under permutation is Proposition 35. Nothing needs to be associative because nothing is composed twice; factorisation of an $n$-ary evaluation into lower ones then becomes a property that holds exactly on causally independent components, rather than an axiom.

*Remark* 40 (What the sum is). FORK. Under the reading in which parallel composition is addition, the formal sum of Definition 34 is available in the calculus: it is the parallel composition of the terminal states. That reading is attractive, since it needs no new former and no tagging, and multiplicity is then counted by $\mid$ itself.

It has a hazard that must be checked before it is adopted. Inertness is not preserved by parallel composition: $x!A$ and $\mathsf{for}(y \leftarrow x)P$ are each inert and their composition is not. So the parallel composition of the summands may reduce further, and the sum would not be a normal form. The alternative is an external formal sum over the multiset of terminal states, which is inert by fiat and costs a coefficient layer outside the calculus.

Two things are settled regardless. Tagging the summands with fresh names is *not* the way to defeat idempotence: a retained tag is a which-path record, distinguishing summands permanently and preventing exactly the coincidence of outcomes on which the addition of coefficients depends. And quotation carries $\mid$ to the name product, so quoting an internalised sum yields a *product* of quotes; whichever reading is taken, the sum must not be internalised beneath a quotation.

*Remark* 41 (Partiality). $\odot$ is partial and the sum may be infinite. Non-confluence is not itself an obstacle — the sum is over all terminal states, not over a chosen one — but divergence is: a composite with no terminal state contributes the empty sum, and a composite with infinitely many contributes a sum that requires a summability condition before it denotes anything. Both are recorded as Obligation 2.

### 8.2 The pole, and residuation

**Definition 42** (Residual)**.** For formulae $\varphi, \psi$ of the generated spatial logic, $\varphi \rhd \psi$ is the right adjoint to the separating conjunction:

$$P \models \varphi \rhd \psi \quad \text{iff} \quad \forall R.\, R \models \varphi \;\Rightarrow\; P \mid R \models \psi.$$

**Definition 43** (The pole)**.** $\perp$ is the behavioural formula

$$P \models \perp \quad \text{iff} \quad \Downarrow P \equiv \mathsf{0},$$

that is, every terminal state of $P$ is the zero process.

**Definition 44** (Linear perp)**.** $P^{\perp} := \{\, Q \;:\; P \odot Q \equiv \mathsf{0} \,\}$, and for a set $S$ of processes, $S^{\perp} := \bigcap_{P \in S} P^{\perp}$.

**Proposition 45** ($P^{\perp}$ is residuation to $\perp$)**.** *$Q \in P^{\perp}$ iff $Q \models \psi_P \rhd \perp$, where $\psi_P$ is a formula whose extension is the equivalence class of $P$ in the logic.*

*Proof.* By Definition 42, $Q \models \psi_P \rhd \perp$ iff $Q \mid R \models \perp$ for every $R$ in the extension of $\psi_P$; by Definition 43 that is $\Downarrow (Q \mid R) \equiv \mathsf{0}$, which is $R \odot Q \equiv \mathsf{0}$ by Definitions 34 and 36. ☐

**Corollary 46** (Symmetry)**.** *$Q \in P^{\perp}$ iff $P \in Q^{\perp}$.*

**Definition 47** (States)**.** $P$ is a *state* when $\{P\}^{\perp\perp} = \{P\}^{\perp\perp}$ is generated by $P$, that is, when $P$ is $\perp\perp$-closed. The states are the carrier of everything that follows.

*Remark* 48 (The closure is the null-space quotient)*.* $\{P\}^{\perp\perp}$ is the set of processes annihilated by every annihilator of $P$, so biorthogonal closure is closure under the testing preorder induced by the pole, with the tests being annihilators and success being reduction to the zero process. Three consequences.

It supplies the collapse that quotation cannot. Quotation is injective on $\mathsf{Proc}/\equiv$ and therefore identifies nothing; every identification in the eventual pairing comes from $\odot$ and from this closure, that is, from reduction.

It fixes the equivalence at which the construction lives, and the equivalence it fixes is *testing*, not bisimulation. Since $\equiv_N$ is intensional and testing is not, the coefficient object and the state space sit at different equivalences; that is a known tension and it is not resolved here.

It destroys freeness. $\mathsf{Proc}/\equiv$ under $\mid$ is free commutative on the non-parallel components, and closed sets are not freely generated. The occupation basis is a coordinate system for the unclosed calculus and does not survive into the closed one.

*Remark* 49 (Totality fails, and how it is repaired). Not every process has an annihilator: a persistent or replicating process at a name no one else holds has none, so $\psi_P \rhd \perp$ is unsatisfiable and the dagger is partial. Definition 47 is the repair adopted here — restrict to the closed elements, which is the standard move — and it is the reason the states are cut down rather than the pole loosened. The alternative, kept in reserve, is to relativise the pole to a residue, taking $R \vdash P \perp Q$ when every terminal state of $P \mid Q$ is $R$; that would make the pairing residue-valued in exactly the way the intended semantics wants, at the cost of a family of poles rather than one.

*Remark* 50 (Canonical witnesses are forced)*.* $P^{\perp}$ is a set, and the dagger must be a term, because it is going to be quoted. Quotation does not respect any behavioural equivalence, so a representative must be chosen and not merely existentially quantified. The choice adopted is by minimal structural complexity, the measure already present for names and processes, with ties broken by the summation of Definition 34 when the minimum is not unique.

### 8.3 The pairing

**Definition 51** (Dagger and pairing)**.** For a state $P$, $P^\dagger$ is the canonical witness of $\psi_P \rhd \bot$ in the sense of Remark 50, composed with the sector and polarity correction of Section 9. The pairing is

$$\langle P \mid Q\rangle \;:=\; @_l\langle P^\dagger \odot Q, \mathsf{0}\rangle$$

and, for a reified operator, using Convention 39,

$$\langle P \mid K \mid Q\rangle \;:=\; @_l\langle\odot\Big(P^\dagger,\, K_x^*,\, x!\langle Q, \mathsf{0}\rangle\Big)\,, \mathsf{0}\rangle$$

with $x$ private to the composite and $K_x^* := \mathsf{for}(y \leftarrow x)K[*_l y]$ the abstraction corresponding to $K$.

*Remark* 52 (Why the quotation is outermost, and why the operator is reified)*.* Two points, each answering an earlier error.

$\odot$ lands in $\mathsf{Proc}$ and the scalars are names, so the pairing must pass through a quotation to change sort; a definition that has $\odot$ produce a scalar directly collapses the two sorts at precisely the place where the collapse is dangerous. The quotation is applied once, after evaluation, and contributes no identifications of its own.

Reifying $K$ removes the bracketing question rather than answering it. Filling a hole and then evaluating, or evaluating and then filling, differ by the cross-redexes of Remark 38; passing the argument along a private wire means there is no hole-filling anywhere, and the bracket is manifestly the three-point evaluation it should have been. Privacy of $x$ is required, not cosmetic: it is what prevents $P^\dagger$ from intercepting the operator's wire, and renaming cannot supply it in a calculus where any party may construct any quotation.

*Remark* 53 (Annihilation is normalisation, not orthogonality)*.* An immediate and initially unwelcome consequence. $\langle P \mid P\rangle = @_l\langle \mathsf{0}, \mathsf{0}\rangle$, the unit of the name product, so annihilation gives 1. The pole is a *normalisation* condition: $P^\dagger$ is the process that reduces $P$ to the unit, and $\langle P \mid P\rangle = 1$ is a theorem rather than a stipulation. That is a gain, but it vacates the position orthogonality was expected to occupy, and the correspondence table of the earlier work, which reads "orthogonality is process annihilation", is backwards on this point.

Orthogonality must therefore come from cancellation among the summands of Definition 34, which requires additive inverses on those summands. This is precisely what the pair sort supplies and what the unexpanded namespace of Section 2 could not: two histories reaching the same terminal state with opposite polarity cancel, and $\langle P \mid Q\rangle = 0$ is the vanishing of a sum of nonzero contributions rather than the absence of any. The expansion of the namespace and the definition of the pairing are answerable to each other at exactly this point.

## 9 The dagger, in one paragraph

The construction this calculus serves requires an antilinear involution on states. Section 8 supplies the pole and the residuation; what remains is the sector and polarity content, recorded here.

First, the dagger is not a syntactic polarity swap on term formers. It is obtained by residuation against a pole: $Q^\dagger$ is a witness of $\psi_Q \rhd \bot$, in the sense made precise by Proposition 45 and Remark 50. This makes the dagger a property first and a term second, hence stable under any presentation generating the same logic. The choice of $\perp$ as the zero process is the simplest available and is provisional; Remark 49 records the relativised alternative.

Second, the sector and polarity components of the dagger are now fixed. Flipping both sectors is the map $(a, b) \mapsto (b, a)$, which in the complex numbers is $z \mapsto i \cdot \bar{z}$ and not conjugation; taken as the dagger it makes the pairing skew-Hermitian and puts $\langle P \mid P \rangle$ in the imaginary sector, so positivity fails outright. The correction is to compose the sector flip with a pair exchange,

$$@_l \mapsto \ominus @_r, \qquad @_r \mapsto \ominus @_l,$$

which is $z \mapsto -i\bar{z}$, is involutive, and is assembled entirely from operations Definitions 5 and 14 already provide. Since Proposition 7 lifts $\ominus$ to processes, this composite acts on a whole process rather than name by name, which is what the pairing of Definition 51 requires and what an earlier presentation could not write down. Remark 8 remains in force: $\ominus$ supplies the dagger's polarity component and is not itself the dagger.

*Remark* 54 (The positivity obligation, made precise)*.* Since every communication rotates the sector and the dagger rotates it once more, the sector in which $\langle P \mid P \rangle$ lands depends on the number of communications consumed in the annihilation of $P^\dagger$ against $P$, modulo four. Positivity therefore requires that all annihilation paths of $P^\dagger \mid P$ agree modulo four. This is a graded confluence condition, weaker than confluence, and it is not obviously true. It is cheap to test on the ladder of mutually annihilating names, and it should be tested before anything further is built on the dagger.

## 10 Obligations

**Obligation 1** (Cross-redexes and the unit of evaluation)**.** Remark 38 makes $\odot$ non-associative, with the deficit $[T, M_R]$. Convention 39 sidesteps the question for a single evaluation but does not answer it: composing operators, $\langle P \mid K_1 K_2 \mid Q \rangle$, reintroduces it in the form of whether the wires create cross-redexes. Determine the condition under which the $n$-ary evaluation factors, and check it against the reified operators of Definition 51.

**Obligation 2** (Summability)**.** Discharge Remark 41: give the condition under which the sum of Definition 34 denotes when the set of terminal states is infinite. Over a positive codomain monotone convergence suffices; with the pair sort supplying signs it does not, and an absolute-summability condition on the annihilation profile is the candidate.

**Obligation 3** (The sum, internal or external)**.** Settle Remark 40. If the sum is parallel composition, show that the composite of the terminal states is inert, or characterise when it is not. If external, say where the coefficient layer lives and how it interacts with Remark 20.

**Obligation 4** (Matching the logic to the pole)**.** Proposition 45 uses a formula $\psi_P$ whose extension is the equivalence class of $P$. If that logic separates more finely than $\perp\perp$-equivalence, then $P \mapsto P^\dagger$ does not factor through the states of Definition 47 and the dagger is ill defined on them. Either take the characteristic formula in a logic whose adequacy target is the pole's equivalence, or define the dagger on terms and prove separately that it descends.

**Obligation 5** (The four-case table)**.** Settle whether the deposited sector is computed from the receipt's channel alone (Definition 23) or from both participants (Section 6.3). Remark 32 is the strongest argument for the latter. This decision governs everything downstream and should be taken first.

**Obligation 6** (Arithmetic or coloured names)**.** Settle Remark 20: whether the completion equations hold on names, and hence whether cancellation is a structural event decided by the matcher or an external one deferred to a coefficient layer.

**Obligation 7** (Decidability of name matching)**.** COMM fires on $\equiv_N$. Under the arithmetic reading of Remark 20, the matcher must normalise formal differences nested through quotation. Termination is not obvious, and this is the practical gate on the entire design.

**Obligation 8** (Freshness)**.** The property that a quotation of $P$ is not free in $P$ is what obviates a freshness operator and underwrites freshness by quotation. Its proof proceeds by structural complexity. Both quotation operators must increase that measure, which is routine; but the quote-drop law of Definition 19 relates a name to a term built from its own drops, and the completion equations, if adopted, shorten terms non-structurally. The mutual recursion of $\equiv$ and $\equiv_N$ was already the delicate part of the metatheory. That argument must be redone rather than adapted.

**Obligation 9** (Observability of the sector)**.** $*_l$ and $*_r$ are term formers, so any context may apply both to a name and see which yields `0`. The sector is therefore classically readable, which makes it a superselection rule rather than a phase, and makes global phase detectable by a two-line context. The natural remedy is to admit the drops in guard position only, where evaluation is pure and non-consuming, and to let term-position dropping be $*_l$ by convention. This needs deciding, and it interacts with Definition 11.

**Obligation 10** (Scalar extraction and occurrence counts)**.** For a race to interfere, its candidates must reach the same outcome with different coefficients. Consuming from $@_l A$ rather than $@_r A$ yields $P\{@_r Q/y\}$ rather than $P\{@_l Q/y\}$, and these are the same outcome up to phase only if the sector can be extracted from the name to the term, that is, only if there is a homomorphism from processes to $\mathbb{Z}/4$ under which substitution multiplies. The natural candidate is sector degree, the product of sectors over name occurrences — but then the phase is raised to the power of the number of occurrences of $y$, so extraction holds only when $y$ occurs exactly once. Occurrence count is the exponent on the phase. Either linear use of names bound from graded channels is required, or multiplicative phase in occurrences is accepted and scalar extraction abandoned. This is a linearity requirement of a different kind from the ones previously considered and declined, and the earlier reasons for declining do not apply to it.

**Obligation 11** (Positivity)**.** Discharge or refute Remark 54: that all annihilation paths of $P^\dagger \mid P$ agree in length modulo four.

**Obligation 12** (Two gradings on names)**.** This calculus grades names by sector. A separate line of work grades them by binding structure, unforgeable names being those introduced by a name-restriction operator and classical ones being quotations. The two gradings are independent, so there are four kinds of name, and their interaction is unstated. The conjecture worth testing is that sectors should live only in the unforgeable namespace, which would also confine Obligation 9.

**Obligation 13** (Metatheory)**.** Nothing in this note is proved beyond Propositions 6, 10 and 15. Subject reduction for the sector grading, whether bisimulation is a congruence in the presence of two quotations, and whether the sector is visible to bisimulation at all, are all open. The last is pressing: an inert distinction that $\equiv$ can see and $\sim$ cannot is exactly the configuration in which the spatial layer strictly refines behaviour, and the sign structure would then be behaviourally invisible.

## 11 What this note does not do

It does not give the interpretation map, and it does not establish that the resulting scalars form a rig — only that the four obstructions to their doing so have each been given a place to be repaired. The evaluation and the pole of Section 8 are stated, not developed: nothing is proved about $\odot$ beyond commutativity, the pole is the simplest one available and is provisional, and the summability question of Obligation 2 is open, so the pairing of Definition 51 is a definition and not yet a well-defined operation. It does not treat the join, whose generalisation to graded messages requires a sum over matchings with coefficients in a group ring, and whose normalisation may or may not remain the right unit. And it takes no position on whether the resulting calculus is worth the cost of Obligation 7, which is the question a reader should keep in view throughout.

## References

[1] L.G. Meredith and M. Radestock. *A reflective higher-order calculus.* Electronic Notes in Theoretical Computer Science, 141(5), 2005.

[2] R. Milner. *Communicating and Mobile Systems: the Pi-Calculus.* Cambridge University Press, 1999.

[3] L. Caires and L. Cardelli. *A spatial logic for concurrency.* Information and Computation, 186(2), 2003.

[4] L. Caires. *Behavioral and spatial observations in a logic for the pi-calculus.* In Foundations of Software Science and Computation Structures, 2004.

[5] J.-Y. Girard. *Linear logic.* Theoretical Computer Science, 50(1), 1987.

[6] S. Abramsky and B. Coecke. *A categorical semantics of quantum protocols.* In Logic in Computer Science, 2004.

[7] B. Coecke and A. Kissinger. *Picturing Quantum Processes.* Cambridge University Press, 2017.

[8] J. Leifer and R. Milner. *Deriving bisimulation congruences for reactive systems.* In CONCUR, 2000.

[9] P. Selinger and B. Valiron. *A lambda calculus for quantum computation with classical control.* Mathematical Structures in Computer Science, 16(3), 2006.

[10] J. Baez. *The octonions.* Bulletin of the American Mathematical Society, 39(2), 2002.